www.csvtuonline.com

337612 (37)

BE (6th Semester)

Examination, Nov.-Dec., 2013

Branch : Mechanical

TURBO MACHINERY

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) and part (b) of each question is compulsory. Solve any one from part (c) and part (d). Write very briefly and to the point Steam tables and Mollier diagram is permitted in the examination.

Q. 1. (a) Why compounding of impulse turbine is necessary ? 2

(b) With suitable sketch briefly explain pressure compounded impulse turbine. 4

17

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

(c) In a De Laval turbine steam issues from the nozzle with a velocity of 1200 m/s. The nozzle angle is 20°, the mean blade velocity is 400 m/s, and inlet and outlet angles of blades are equal. The mass of steam flowing through the turbine per hour is 1000 kg. Calculate :

(i) Blade angles

(ii) Relative velocity of steam entering the blades

(iii) Tangential force on the blades

(iv) Power developed

(v) Blade efficiency

Take blade velocity coefficient as 0.8 10

Or

(d) What should be optimum value of blade to steam velocity and hence derive expression

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

for maximum efficiency in a single stage impulse turbine.

Q. 2. (a) Define degree of reaction. 2

(b) What are the different types of internal losses in turbine ? 4

(c) At a particular stage of a reaction turbine, the mean blade speed is 60 m/s and the steam pressure is 3.5 bar with a temperature of 175°C. The identical fixed and moving blades have inlet angles of 30° and outlet angles of 20°. Determine :

(i) the blade height if it is 1/10 of the blade ring diameter, for a flow rate of 13.5 kg/s

(ii) the power developed by a pair and

(iii) the specific enthalpy drop if the stage efficiency is 85%. 10

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

Or

(d) In a reaction turbine, the blade tips are inclined at 35° and 20° in the direction of motion. The guide blades are of the same shape as the moving blades but reversed in direction. At a certain place in the turbine, the drum diameter is 1 m and blades are 10 cm high. At this place, the steam has a pressure of 1.75 bar and 0.935 dry. If the speed of the turbine is 2500 rpm and the steam passes through the blades without shock, find the weight of steam flow and power developed in the ring of the moving blades.

Q. 3. (a) Define reheat factor. 2

(b) What is throttle governing. (Give suitable sketch) 4

www.csvtuonline.com

www.csvtuonline.com

www.csvtuonline.com

each turbine in order that the H.P. and I.P. ma

each developed 1/4 of the total power. Assum

stage efficiencies of 0.77, 0.75 and 0.72 in th

H.P., I.P. and L.P. respectively.

Q. 4. (a) Draw T-s diagram of Brayton cycle.

(b) Draw the schematic and T-s diagram of ga

turbine plant with regeneration.

(c) What is the optimum pressure ratio f

maximum specific work output of a gas turbi

plant. 1

Or

(d) The air enters the compressor of an ope

cycle constant pressure gas turbine at

pressure of 1 bar and temperature of 20°C

The pressure of the air after compression is

bar The isentropic efficiencies of th

compressor and turbine are 80% and 85%

www.csvtuonline.com

H.P. and I.P.

al power. Assu

75 and 0.72

y.

cycle.

diagram of g

n.

ssure ratio

t of a gas turb

sor of an o

as turbine a

erature of 2

ompression

encies of

80% and

www.csvtuonline.com

respectively. The air fuel ratio used is 90 : 1. If flow rate of air is 3.0 kg/s, find

(i) power developed

(ii) thermal efficiency of the cycle.

Assume Cp = 1.0 kJ/kg-K and γ = 1.4 of air and gases. Calorific value of fuel = 41800 kJ/kg.

Q. 5. (a) Define slip factor. 2

(b) Define surging and choking. 4

(c) A centrifugal compressor running at 9000 rpm delivers 600 m^3/min of free air. The air is compressed from 1 bar and 20°C to a pressure ratio of 4 with an isentropic efficiency of 0.82. Blades are radial at outlet of impeller and flow velocity of 62 m/s may be assumed throughout constant. The outer radius of impeller is twice the inner and the slip factor may be assumed as 0.9. The blade area

20

www.csvtuonline.com

www.csvtuonline.com

coefficient of 0.9 may be assumed at inlet.

Calculate the following :

(i) Final temperature of air

(ii) Theoretical power

(iii) Impeller diameters at inlet and outlet

(iv) Breadth of impeller at inlet

(v) Impeller blade angle at inlet

(vi) Diffuser blade angle at inlet 10

Or

(d) An axial flow compressor, with compression ratio as 5, draws air at 20°C delivers it at 50°C. Assuming 50% degree of reaction, find the velocity of flow if the blade velocity is 100 m/s. Also find the number of stages. Take work factor = 0.85, $\alpha = 10°$, $\beta = 40°$ and Cp = 1 kJ/kg K.

www.csvtuonline.com

www.csvtuonline.com